「Lemma.

Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F. Then,

For any $\sigma \in \mathrm{Aut}(K/F)$, $\sigma\alpha$ is a root of the minimal polynomial for α over F.

Proof. Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in F[x]$ be the minimal polynomial for α.
Then, $f(\alpha)=0$, so $\sigma(f(\alpha))=0$. (The field Homomorphism preserves zero).
Now,

$$
\begin{aligned}
0 = \sigma(f(\alpha)) &= \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0) \\
&= (\sigma\alpha)^n + \sigma(a_{n-1})\cdot(\sigma\alpha)^{n-1} + \cdots + \sigma(a_1)\cdot(\sigma\alpha) + \sigma(a_0) \\
\sigma \text{ fixes } F \rightarrow\ &= (\sigma\alpha)^n + a_{n-1}(\sigma\alpha)^{n-1} + \cdots + a_1\cdot(\sigma\alpha) + a_0 \\
&= f(\sigma\alpha).
\end{aligned}
$$

「Prop. Let $H \le \mathrm{Aut}(K)$ be a subgroup of the group of all automorphisms of K.
Then, the collection F of elements of K fixed by all the elements of H is a subfield.

Proof. First, given $\sigma \in \mathrm{Aut}(K)$, the set $F_\sigma = \{a \in K \mid \sigma(a) = a\}$ is subfield, because:
Let $a, b \in F_\sigma$ be given. Then, $\sigma(a-b) = \sigma(a) - \sigma(b) = a-b$, and if $b \neq 0$,
then $\sigma(a\cdot b^{-1}) = \sigma(a)\cdot\sigma(b)^{-1} = ab^{-1}$, so the claim is proved.
And now, since the arbitrary intersection of fields is field, So Proved. ☒

Such a subfield is called the fixed field of $S \subseteq \mathrm{Aut}(K)$.

「Prop. Let K be a field. Then,

If $F_1 \le F_2 \le K$, then $\mathrm{Aut}(K/F_2) \le \mathrm{Aut}(K/F_1)$.

If $H_1 \le H_2 \le \mathrm{Aut}(K)$, then $F_{H_2} \le F_{H_1}$.

In other say, The inclusion relation of automorphisms and fixed field is reverse.

Lemma.

Def. A finite extension K/F is called a Galois Extension if:

$$|Aut(K/F)| = [K:F].$$

In such a case, denote $Gal(K/F) = Aut(K/F)$.

Def. A character χ of a group G with values in a field L is a group Homomorphism

$$\chi : G \to L^\times$$

The characters $\chi_1, \cdots, \chi_n$ of G are said to be linearly independent over L if:

$a_1\chi_1 + \cdots + a_n\chi_n \equiv 0$ for some $a_i \in L$, then $a_1 = \cdots = a_n = 0$.

Thm. [DeDekind - Independence of Characters Theorem].

If $\chi_1, \cdots, \chi_n : G \to L^\times$ are distinct characters of G with values in L, then they are linearly independent over L.

Proof. Suppose that the characters are linearly dependent.

That is, there exists non-trivial representation; for some $a_1, \cdots, a_n \in L$,

$$a_1\chi_1 + \cdots + a_n\chi_n \equiv 0 \text{ where } a_1, \cdots, a_n \text{ are not all zero.}$$

So, the set $\{m \in \mathbb{N} \mid \text{for some } b_1, \cdots, b_m \in L\setminus\{0\}, \text{ and } \sigma \in S_n, b_1\chi_{\sigma(1)} + \cdots + b_m\chi_{\sigma(m)} \equiv 0\}$ is non-empty, thus it has a minimal element by the well-ordering principle,

$m \in \mathbb{N}$ with $b_1\chi_1 + \cdots + b_m\chi_m \equiv 0$, the indices are renumbered, $b_i \in L\setminus\{0\}$.

Since the characters are distinct, $\chi_1(g_0) \neq \chi_m(g_0)$, for some $g_0 \in G$.

Given $g \in G$, observe that

$$0 = b_1\chi_1(g_0 g) + \cdots + b_m\chi_m(g_0 g) = b_1\chi_1(g_0)\chi_1(g) + \cdots + b_m\chi_m(g_0)\chi_m(g)$$

Meanwhile,

$$0 = \chi_m(g_0)\cdot 0 = \chi_m(g_0)\cdot(b_1\chi_1(g) + \cdots + b_m\chi_m(g)) = b_1\chi_m(g_0)\chi_1(g) + \cdots + b_m\chi_m(g_0)\chi_m(g),$$

Substracting gives $[\chi_m(g_0) - \chi_1(g_0)]\cdot b_1\chi_1(g) + \cdots + [\chi_m(g_0) - \chi_{m-1}(g_0)]\cdot b_{m-1}\chi_{m-1}(g_0) = 0$, because the the last term is terminated. But, since $[\chi_m(g_0) - \chi_1(g_0)] \neq 0$, it is another representation, but that is contradiction to the minimality of m. ∎

Corollary. Distinct Field Automorphisms are linearly independent.

「Thm.

Let $G = \{\sigma_1 (= id), \sigma_2, \cdots, \sigma_n\}$ be a subgroup of $Aut(K)$, and let $F = K^G$.
Then, $[K:F] = |G| = n.$

Proof. To show the equality, we will show that neither $|G| > [K:F]$ nor $|G| < [K:F]$.
Suppose that $n > [K:F]$ and let $w_1, \cdots, w_m$ be a basis for K over F. (So, $m = [K:F]$)
Consider a system

$$\begin{aligned} \sigma_1(w_1)\cdot x_1 + \sigma_2(w_1)\cdot x_2 + \cdots + \sigma_n(w_1)x_n &= 0 \\ &\vdots \\ \sigma_1(w_m)x_1 + \sigma_2(w_m)x_2 + \cdots + \sigma_n(w_m)x_n &= 0 \end{aligned} \quad \left(\text{or } \left(\sigma_j(w_i)\right)_{ij} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0 \right)$$

By the assumption, $n > m$, so the system has a non-trivial solution $\beta_1, \cdots, \beta_n$ in K.
(By the dimension theorem, nullity $A = n - \text{rank } A \geq n - m > 0$.)
Given $a_1, \cdots, a_m \in F$, these are fixed by $\sigma_i \in G$, by the definition of F.
Multiflying a_i to the ith row of the equation, then

$$\begin{aligned} \sigma_1(a_1 w_1)\cdot \beta_1 + \cdots + \sigma_n(a_1 w_1)\beta_n &= 0 \\ &\vdots \\ \sigma_1(a_n w_n)\beta_1 + \cdots + \sigma_n(a_n w_n)\beta_n &= 0. \end{aligned}$$

So, adding all the equations; for $\alpha = a_1 w_1 + \cdots + a_n w_n$,

$$\beta_1 \sigma_1(\alpha) + \cdots + \beta_n \sigma_n(\alpha) = 0.$$

Since $a_1, \cdots, a_m$ was arbitrary and $w_1, \cdots, w_m$ forms basis for K over F,
the α is arbitrary element of K. Moreover, since $\beta_1, \cdots, \beta_n$ are not all zero,
it contradicts the G is linearly independent, by the dedekind Character Theorem.」
Now, suppose that $n < [K:F]$. That is, there are more than n F-linearly independent
elements, say $\alpha_1, \cdots, \alpha_{n+1}$. Consider a system of $n+1$ unknowns,

$$\begin{aligned} \sigma_1(\alpha_1)x_1 + \cdots + \sigma_1(\alpha_{n+1})x_{n+1} &= 0 \\ &\vdots \\ \sigma_n(\alpha_1)x_1 + \cdots + \sigma_n(\alpha_{n+1})x_{n+1} &= 0. \end{aligned}$$

Then, it has n equation and $n+1$ unknowns
So there is non-trivial solution $\beta_1, \cdots, \beta_{n+1}$ in K.
Moreover, since $\sigma_1 = id$ and $\alpha_1, \cdots, \alpha_{n+1}$ are F-linearly independent, not all β_i lies in F.

Corollary. Let K/F be any finite extension. Then,

$$|\mathrm{Aut}(K/F)| \leq [K:F]$$

In Particular, the equality holds if and only if $F = K^{\mathrm{Aut}(K/F)}$

if and only if K/F is Galois extension

if and only if K is splitting field of separable polynomial over F.

Proof.

「Corollary.

Let G be a finite subgroup of $\mathrm{Aut}(K)$. Then

$$\mathrm{Aut}(K/K^G) = G.$$

And, if $G_1 \neq G_2$, then $K^{G_1} \neq K^{G_2}$.

「Fundamental Theorem of Galois Theory.

Let K/F be a Galois Extension. Denote $G = Gal(K/F)$.

1). A map

$$\varphi : \{ H \leq Gal(K/F) \mid H \text{ is subgroup} \} \to \{ F \leq E \leq K \mid E \text{ is intermidate field} \}$$
$$: H \mapsto K^H$$

is bijection, with the inverse $\varphi^{-1}(E) = Gal(K/E)$.

2). Given $H_1, H_2 \leq G$,

$$H_1 \leq H_2 \text{ if and only if } K^{H_2} \leq K^{H_1}.$$

3). Given $H \leq G$,

$$[K : K^H] = |H| \quad \text{and} \quad [K^H : F] = |Gal(K/F) : H|$$

$$\begin{tikzcd} K \arrow[d, dash, "|H|"] \arrow[dd, dash, bend right, "|G|"'] \\ K^H \arrow[d, dash, "|G:H|"] \\ F \end{tikzcd}$$

4). Given $H \leq G$, K/K^H is always Galois, with $Gal(K/K^H) = H$.

Moreover, K^H/F is Galois if and only if $H \trianglelefteq G$ is normal.

If K^H/F is Galois, then $Gal(K^H/F) \cong Gal(K/F)/H$.

Further, there exists one-to-one correspondence between

$$\{ \varphi : K^H \to \overline{K} \, ; \, \varphi \text{ is embedding which fix } F \} \leftrightarrow \{ \sigma H \mid \sigma \in G \},$$

5). Given $H_1, H_2 \leq G$, then

$$K^{\langle H_1, H_2 \rangle} = K^{H_1} \cap K^{H_2} \quad \text{and} \quad K^{H_1 \cap H_2} = K^{H_1} K^{H_2}$$